

APA NEAGRA, Romania

WMO: 153410

Lat: 44.9969N

Lon: 22.8594E

Elev: 823

StdP: 14.26

Time Zone: 2.00 (EUE)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.4	13.1	5.9	7.7	11.9	10.5	9.7	15.7	18.7	43.5	14.2	40.0	0.9	10	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.2	89.5	70.6	87.0	70.4	84.4	69.9	76.0	83.4	73.5	82.1	71.6	80.4	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.8	130.0	79.3	70.9	117.2	77.3	68.8	108.9	75.5	40.0	81.8	37.6	82.1	35.8	80.5	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	12.9	9.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	28.4	35.0	42.3	51.6	60.6	67.6	69.4	70.8	61.5	50.5	41.5	33.2		
	DBStd	15.73	7.73	7.52	7.53	7.26	5.06	4.98	4.69	4.93	6.61	6.78	7.37	6.79		
	HDD50	2278	669	420	254	68	1	0	0	0	3	78	265	520		
	HDD65	5569	1134	839	704	404	150	30	16	12	141	449	705	985		
	CDD50	2683	0	1	15	115	330	528	600	643	348	93	10	0		
	CDD65	500	0	0	0	0	15	108	151	190	36	0	0	0		
	CDH74	5885	0	0	1	63	226	1186	1767	2106	500	36	0	0		
	CDH80	1884	0	0	0	4	21	335	574	832	117	1	0	0		
(14) Wind	WSAvg	2.5	1.6	2.3	2.8	3.6	2.8	2.7	2.9	2.6	2.6	2.5	1.8	2.0		
(15) Precipitation	PrecAvg	31.9	2.1	1.9	1.9	2.8	3.2	3.4	3.5	1.9	3.0	2.8	2.6	3.1		
	PrecMax	53.7	4.3	6.2	4.4	6.5	7.0	6.2	11.8	4.4	5.7	6.2	7.1	8.3		
	PrecMin	13.5	0.2	0.1	0.1	0.6	0.8	0.5	0.4	0.2	0.2	0.0	0.1	0.1		
	PrecStd	8.8	1.3	1.4	1.3	1.3	1.5	1.8	2.5	1.2	1.9	1.8	1.8	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	60.8	72.6	79.9	82.7	90.7	91.7	93.8	87.4	78.0	66.2	56.7		
		MCWB	45.6	48.8	54.3	60.9	65.6	71.6	72.2	72.8	66.8	60.4	55.3	47.8		
	2%	DB	47.8	55.4	67.4	75.4	79.1	86.7	88.4	90.3	83.3	73.6	59.9	52.2		
		MCWB	42.1	46.9	52.3	60.4	65.5	71.1	70.6	70.6	66.6	59.3	52.9	44.8		
	5%	DB	42.6	50.6	61.1	71.2	76.5	83.9	85.9	87.8	79.8	68.7	55.6	47.5		
		MCWB	38.1	44.3	49.3	57.7	64.4	70.3	70.4	70.4	64.8	57.9	51.6	42.9		
	10%	DB	39.0	46.8	56.3	66.5	73.5	80.8	83.2	84.6	75.9	63.9	52.1	43.2		
		MCWB	35.8	42.2	46.8	55.2	63.4	68.3	69.4	70.2	63.9	55.8	49.6	39.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.2	52.1	57.0	64.3	71.7	78.2	78.4	82.0	74.4	64.9	57.4	49.1		
		MCDB	52.4	56.6	70.7	74.4	75.7	83.9	83.5	85.1	82.8	68.2	63.7	52.6		
	2%	WB	42.5	48.3	53.4	61.6	68.2	73.7	75.8	75.9	70.5	61.2	54.6	45.9		
		MCDB	46.7	53.2	64.0	73.0	75.1	83.0	81.8	85.0	76.1	69.3	58.0	51.3		
	5%	WB	39.2	45.6	50.7	59.3	65.7	71.7	73.1	73.0	67.6	59.2	52.5	43.2		
		MCDB	41.6	48.8	58.9	70.0	73.1	80.4	81.1	82.5	74.1	67.5	55.3	46.9		
	10%	WB	36.5	42.8	48.3	56.3	63.9	69.7	71.1	71.0	65.3	56.6	50.2	40.7		
		MCDB	38.2	46.5	54.5	64.6	71.6	78.9	80.0	80.7	73.0	62.9	51.9	43.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	18.1	20.9	22.1	20.9	23.0	25.1	25.2	23.4	22.7	15.5	16.6		
		MCDBR	22.2	25.4	31.3	31.0	26.7	27.1	28.3	30.1	30.1	32.9	22.3	23.6		
	5% WB	MCWBR	16.3	17.2	17.3	15.9	13.5	12.6	13.3	12.6	13.2	19.7	14.9	17.0		
		MCWBR	19.3	22.9	27.0	29.7	22.7	24.4	24.4	25.8	23.6	29.2	18.4	22.3		
(40) Clear-Sky Solar Irradiance	taub		0.340	0.376	0.422	0.491	0.471	0.460	0.478	0.500	0.453	0.408	0.374	0.340		
	taud		2.244	2.147	2.101	1.946	2.064	2.145	2.102	2.032	2.117	2.231	2.267	2.279		
	Ebn at Noon		236	248	253	245	255	259	252	240	240	235	221	223		
	Edh at Noon		28	36	44	56	51	47	49	51	43	33	27	24		
(44) All-Sky Solar Radiation	RadAvg		427	679	1050	1423	1717	1930	1914	1715	1221	819	475	355		
	RadStd		57	93	141	156	148	173	143	161	117	104	64	49		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-283	+194	+106

Nomenclature: See separate page